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Does Dealer Gamma Carry Volatility Information Beyond VIX?

Abstract

Dealer gamma tracks realized volatility strongly: short-gamma days carry far higher RV (Welch $t \approx +28$ over 15 years). But almost all of that is already in VIX: a variance decomposition puts 97.8% of gamma’s own explanatory power for log-RV inside a VIX/HAR baseline, leaving a 2.2% incremental sliver. On a calm 21-month options window that residual is undetectable, a clean null across six pre-registered formulations. On 15 years spanning real stress regimes, a small, statistically robust, gamma-specific increment survives: gamma-only Diebold-Mariano on CRPS $p = 0.001$ (Clark-West, the correct test for this nested comparison, agrees at $p < 0.001$), ΔAUC $p = 0.001$. The increment is genuinely gamma, not the DIX flow signal that ships alongside it (adding DIX on top of gamma dilutes rather than helps, $p = 0.006$ vs 0.001), it survives a richer VIX baseline rather than proxying a stale VIX, and it is economically small. Finding it at all took statistical power, multiple regimes, and a confound check.

The same 21-month options window also backs a second, independent signal test with the identical kill-switch protocol: 25-delta put-call skew (§5c). It is not a VIX echo story like gamma, it is a cleaner, more decisive null, both formulations tested significantly underperform the baseline.

Runnable evidence is in [analysis/](#). This is the signal investigation behind one of the inputs to the strategy in [STRATEGY.md](#).

1. The question, and why the obvious framing is wrong

Options dealers hedge their inventory, and the direction of that hedging flips with their gamma position. When dealers are long gamma they hedge counter-trend, which suppresses realized volatility and pins price; when they are short gamma, below the gamma flip, they hedge trend-following, which amplifies moves. So gamma plausibly modulates the volatility regime. The naive test, “does gamma beat VIX at forecasting the level of RV,” is a trap, because VIX is by construction

the market's price of forward variance and will absorb most of any volatility signal. The sharp, falsifiable question is the incremental one:

Does dealer gamma carry RV-regime information incremental to a VIX/HAR baseline, and if so, how large, and where?

A null is a fully acceptable answer to that question. As it turns out the answer is “almost none, but not zero on a powered sample,” and stating that precisely is the point.

2. Data (all free; downloaded and git-ignored; the fetcher ships)

Source	What	Window
SqueezeMetrics DIX.csv	daily dealer GEX (and DIX) for the S&P	2011-05 → 2026-05 (3,791 rows, verified live)
CBOE VIX_History.csv	VIX	1990 → 2026
yfinance	SPY OHLC (Yang-Zhang RV), VIX3M / VIX9D / VVIX	2010/2011 → 2026
Databento OPRA (owned)	signed net gamma and by-strike profile (the 21-month sub-study)	2024-08 → 2026-03
FRED	DGS3MO risk-free	daily

SqueezeMetrics' `gex` is negative on 9.1% of the deep window (~345 short-gamma days, clustered in 2011/2015/2018/2020/2022), which is enough to study the amplification regime that the calm owned window could not see.

3. Method

- Contamination-fixed target: RV regime versus a baseline ending at $t-1$, excluding the present value, so the comparison cannot leak the day it is being measured against.
- Pre-registration of every mechanism-derived formulation, and strict no-lookahead (predictors $\leq t-1$, gamma lagged for OCC's T-1 open interest).
- Out-of-sample expanding walk-forward (~2y initial train, ~3,200 OOS days).
- The right test: a Diebold-Mariano test on the CRPS differential of nested models (VIX/HAR versus +gamma), with Newey-West HAC and the Harvey small-sample correction, alongside a Clark-West test (the textbook correction for DM's conservative bias on nested models); binary targets via OOS log-loss and AUC with a stationary block-bootstrap. DM and the AUC bootstrap are two-sided tests of a one-sided (directional) hypothesis, “gamma helps”; CW is reported one-sided, as is standard for nested-model comparisons.
- Per-regime reporting, never pooled across the 0DTE structural break (pre-2020 / 2020-21 / 2022+).
- Confound decomposition separating gamma from DIX and from stale VIX, with multiplicity in view.

Implementation in pure NumPy/SciPy/scikit-learn: `analysis/phase1_deep_history.py`, `analysis/phase1_robustness.py`, plus the 21-month sub-study `analysis/phase0_gonogo.py`, `phase05_reframe.py`, `phase05b_profile.py`.

4. The mechanism is real and stable across regimes

Mean log realized vol by dealer-gamma sign, deep history:

Era	short-gamma logRV	long-gamma logRV	Welch t
2011–2026 (all)	−1.41 (n=338)	−2.29 (n=3,385)	+27.6
pre-2020	−1.48	−2.38	+20.4
2020–21	−0.99	−2.12	+9.9
2022+	−1.49	−2.18	+16.9

Short-gamma days carry dramatically higher RV in every regime ($p \approx 0$ throughout). Gamma is genuinely informative. The whole question is how much of that survives controlling for VIX.

5. Results

5a. The 21-month owned window: a clean null

On 2024-08 → 2026-03 (415 days, one calm regime), gamma added nothing beyond VIX/HAR across six pre-registered formulations: level, intraday range (pinning), mean-reversion, downside tails, regime-direction, and by-strike profile shape (the profile features actually overfit and degraded the forecast). Underpowered and confined to one regime, this window cannot see a small effect. Details in `analysis/phase0*.py`.

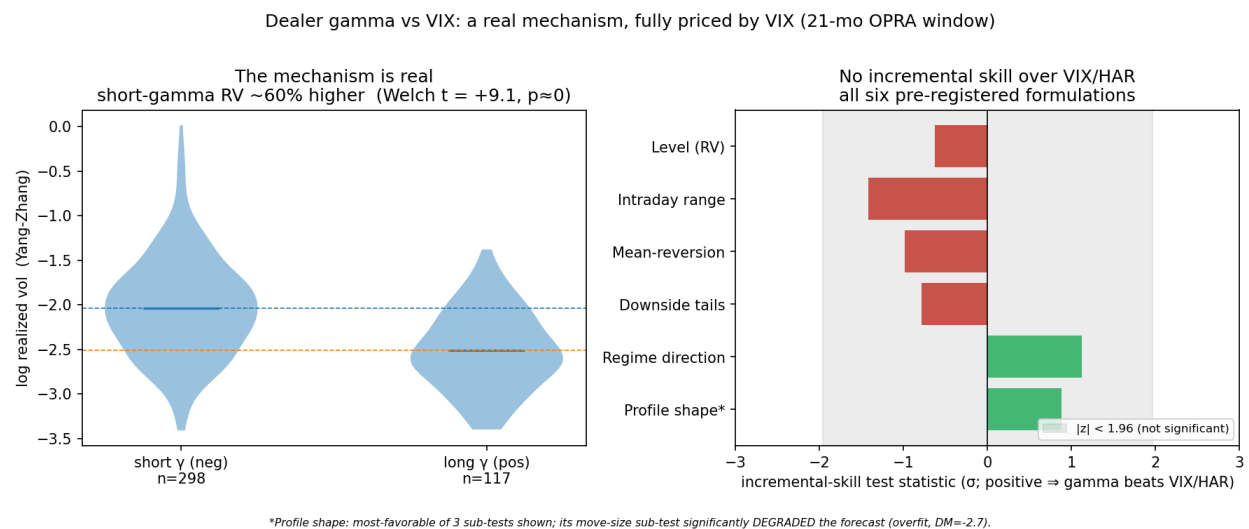


Figure 1: 21-month sub-study: no incremental skill across six formulations

5b. The deep history (2011–2026): a small but robust increment

Incremental skill of gamma over a full VIX/HAR baseline, out-of-sample:

Block	n (OOS)	dCRPS	DM p (2-sided)	CW p (1-sided)	Δ AUC	AUC p	Verdict
All	3,219	+0.0020	0.006	<0.001	+0.007	0.003	gamma helps
pre-2020	1,617	+0.0016	0.115	<0.001	+0.005	0.192	null (positive)
2020–21	498	+0.0022	0.229	0.004	+0.009	0.108	null (positive)
2022+	1,104	+0.0024	0.021	<0.001	+0.007	0.067	gamma helps

DM is the conservative test here: it does not credit the extra gamma parameters unless they clear a high bar, so a DM null (pre-2020, 2020-21) does not mean CW agrees. CW is one-sided by construction (it tests whether the larger model wins) and comes in stronger everywhere, exactly the pattern the DM-vs-CW literature predicts for a nested comparison.

The confound decomposition is the critical check, because SqueezeMetrics ships gamma and DIX (a flow signal, not gamma):

Added to VIX/HAR	DM p (2-sided)	CW p (1-sided)	AUC p
gamma only (gex pct + neg-flag)	0.001	<0.001	0.001
DIX only	0.97 (null)	0.007 (significant)	0.78
gamma, over VIX + Δ VIX baseline	0.001	<0.001	0.001

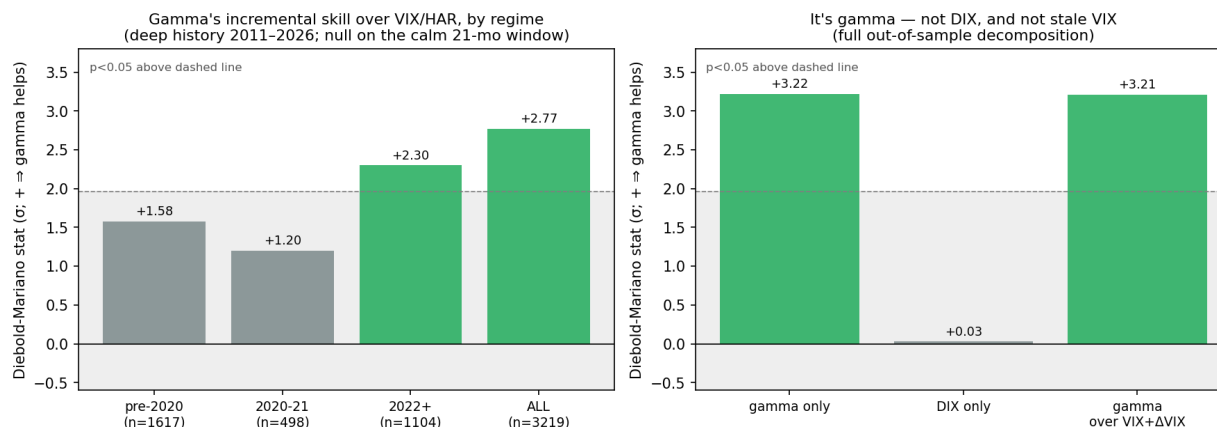
DIX-only is the one case where DM and CW disagree outright: DM calls it a clean null, CW calls it significant. This project does not resolve that disagreement in DIX's favor; both numbers are reported and DIX-only stays an open question rather than a settled null. It does not change the headline conclusion, which rests on DM (the primary test throughout this project) applied to the combination: gamma-only beats gamma+DIX on DM ($p=0.001$ vs 0.006), so adding DIX on top of gamma dilutes rather than helps. CW does not show the same ordering here (its statistic is marginally larger for gamma+DIX than gamma-only), which is expected: CW is known to reward added parameters somewhat regardless of true incremental value, which is exactly why it is used only to check that DM's conservatism is not hiding real gamma signal, not as the test for judging whether an addition helps. On the test built for that judgment, DIX adds nothing once gamma is already in the model.

It is also small: dCRPS $\approx +0.002$ on a baseline CRPS ≈ 0.22 (under 1% relative), Δ AUC $\approx +0.007$. A variance decomposition (`analysis/phase1_robustness.py`, in-sample R^2 of log-RV on gamma alone vs. on VIX/HAR vs. on both) puts 97.8% of gamma's own explanatory power inside VIX/HAR (R^2 0.307 $\rightarrow$ incremental R^2 0.0067 once VIX/HAR is already in the model): gamma is almost entirely a VIX echo, and the last 2.2% is real.

5c. A second options-market signal, same window: put-call skew

The same 21-month OPRA window backs a second, independent kill-switch test, same protocol as §5a: does the market's own risk reversal (25-delta put IV minus 25-delta call IV, `features/skew.py`, walk-forward DM/CRPS vs. HAR/VIX, `analysis/phase_skew.py`) carry next-day RV information beyond VIX/HAR? No, and more decisively than gamma's clean null: both formulations tested significantly underperform the baseline (raw skew dCRPS= -0.00086 , DM $p=0.004$; level-normalized skew dCRPS= -0.00069 , DM $p=0.026$), the same "added complexity without payoff" pattern §5a already found in gamma's own by-strike profile shape. The skew itself is real and persistent even

Dealer gamma carries a SMALL but robust increment beyond VIX (deep, multi-regime, confound-checked)



Effect is economically small: $\Delta CRPS = +0.002$ on baseline $= 0.22$ ($< 1\%$ relative), $\Delta AUC = +0.007$. Gamma is $\sim 95\%$ a VIX echo; the surviving sliver is real. Gamma-only is cleaner than gamma+DIX ($p = 0.001$ vs 0.006).

Figure 2: Deep-history result: a small but robust gamma-specific increment

where it adds no forecasting power: positive (puts richer than calls, the standard equity direction) on 97.3% of days, averaging +5.17 vol points (std 2.89, range -1.8 to $+17.8$).

SPY options: smile, term structure, and skew (OPRA statistics, 2024-08 -> 2026-04)

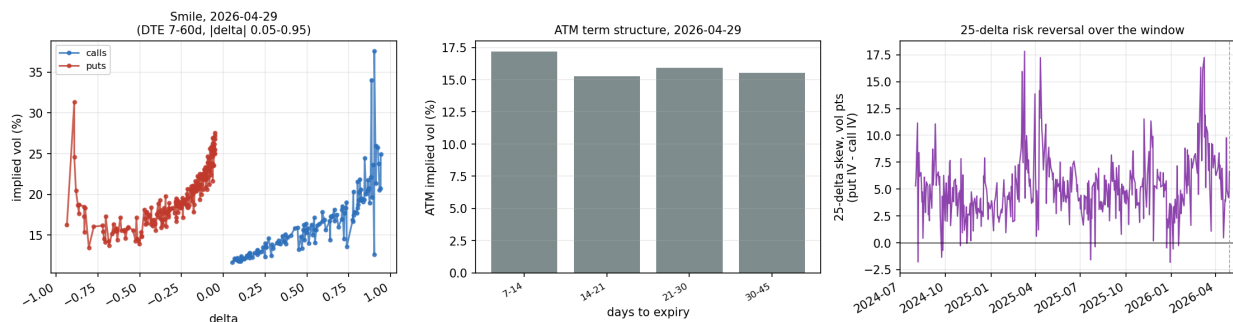


Figure 3: SPY options smile, term structure, and skew, 2024-08 -> 2026-04

The smile figure plots every live contract in the delta-0.05-to-0.95 band on the most recent trading day (2026-04-29) rather than a single curated maturity, so the visible noise at the delta extremes is real thin-tape variance in the underlying statistics-derived closes, not hidden. The 45-60-day-to-expiry bucket is absent from the term-structure panel on this specific day: the monthly-expiry-only universe (`build_id_list.py`'s convention, used everywhere in this project) does not always have a live contract in every DTE bucket on every day.

6. Conclusion

Dealer gamma carries a small, statistically robust increment to next-day RV forecasting beyond VIX. It is detectable only on a powered, multi-regime sample, it is gamma-specific rather than a DIX or stale-VIX artifact, and it was invisible on a calm 21-month window, which is a reminder that “no signal” on a short single-regime sample is a statement about power, not about the world.

Whether a sub-1%-CRPS, 0.7-AUC-point edge is tradeable after costs is a separate question, and the answer in [STRATEGY.md](#) is no: dealer gamma adds nothing to a working short-vol carry once the VIX term structure is already in the model (every gamma overlay tested reduces risk-adjusted return). That is the trading-side corroboration of this document: gamma's increment is real but so small it rounds to zero against VIX. The result rides on SqueezeMetrics' proprietary gamma model, so an independent reconstruction, and the intraday/ODTE timescale where the mechanism is strongest, are the natural next tests.

A first growth probe (`analysis/phase2_learned_flip.py`) finds the daily gamma→RV relationship is a smooth, near-linear gradient in gamma percentile rather than a sharp threshold at the flip. A regime-switching learned-flip model does not beat the plain linear gamma term (DM $p=0.35$), so on daily data the edge is the small linear sliver and a nonlinear model would add complexity without payoff. Any larger, tradeable structure would have to live at the intraday scale, which makes intraday the highest-value next test.

7. Reproduce

```
# env with pandas/numpy/scipy/scikit-learn + pyarrow; fetchers download free data
python analysis/phase1_deep_history.py      # deep-history test, per-regime
python analysis/phase1_robustness.py        # gamma-vs-DIX + richer-VIX decomposition
python analysis/phase0_gonogo.py            # 21-month sub-study (level)
python analysis/phase05_reframe.py          # 21-month sub-study (path/dynamics/tails/regime)
python analysis/phase05b_profile.py         # 21-month sub-study (profile shape)
python -m features.opra_panel               # raw OPRA DBN -> options_panel.parquet (free, local decode)
python -m features.assemble configs/features.yaml # -> features_panel.parquet (GEX + skew)
python analysis/phase_skew.py               # 21-month sub-study (put-call skew, 5c)
python analysis/make_figure_iv_surface.py   # -> analysis/figures/iv_surface.png
```

8. References

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